



Progressive Education Society's
MODERN COLLEGE OF ENGINEERING, Pune -05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
MCA Department

PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA	Div: B	Course Code: MCA10602	Batch: S2
Semester: III		Course Name: Laboratory Practice III	
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CO No: CO607A.3		Assignment No: 07	

Title: Write a program for Building an Image Classification model.

Code:

```
import tensorflow as tf
from tensorflow.keras import datasets, layers, models
import matplotlib.pyplot as plt

(x_train, y_train), (x_test, y_test) = datasets.mnist.load_data()

x_train, x_test = x_train / 255.0, x_test / 255.0

x_train = x_train.reshape((x_train.shape[0], 28, 28, 1))
x_test = x_test.reshape((x_test.shape[0], 28, 28, 1))

model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu', input_shape=(28, 28, 1)),
    layers.MaxPooling2D((2,2)),
    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

history = model.fit(x_train, y_train, epochs=5, validation_data=(x_test, y_test))

test_loss, test_acc = model.evaluate(x_test, y_test)
print(f'\nTest accuracy: {test_acc:.4f}')

plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
```

```
plt.title('Model Accuracy')  
plt.xlabel('Epoch')  
plt.ylabel('Accuracy')  
plt.legend()  
plt.show()
```

Output:

